

MELFORT, SK, Canada

WMO: 714560

Lat: 52.8164N

Lon: 104.5964W

Elev: 490

StdP: 95.58

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.7	-36.5	0.1	-33.1	-33.8	0.2	-30.7	11.0	-9.4	10.0	-9.7	3.5	270	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.6	18.9	26.7	18.1	25.1	17.2	21.1	26.0	19.8	24.3	18.6	23.1	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.1	23.7	18.1	13.8	22.2	16.9	12.8	20.9	63.7	26.1	58.9	24.3	54.9	23.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.9	8.0	-36.6	32.2	2.6	2.0	-38.5	33.6	-40.0	34.7	-41.5	35.8	-43.3	37.3		
			-36.7	23.8	2.6	1.6	-38.6	25.0	-40.1	25.9	-41.6	26.8	-43.4	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-16.4	-14.5	-7.8	2.2	10.0	15.3	17.6	16.6	11.6	3.5	-6.4	-13.8
	DBStd	13.77	8.80	7.76	7.76	6.21	5.08	3.73	3.00	3.53	4.75	5.07	6.97	8.12
	HDD10.0	3845	820	687	553	241	65	4	0	1	36	210	491	738
	HDD18.3	6189	1078	920	812	484	261	104	51	76	206	461	741	996
	CDD10.0	768	0	0	0	7	65	163	234	206	85	7	0	0
	CDD18.3	69	0	0	0	0	3	13	27	23	4	0	0	0
	CDH23.3	792	0	0	0	3	58	147	246	258	79	2	0	0
	CDH26.7	171	0	0	0	1	13	30	42	67	18	0	0	0
Wind	WSAvg	4.1	4.1	4.1	4.2	4.4	4.5	4.1	3.6	3.6	4.1	4.2	4.1	4.0
Precipitation	PrecAvg	416	16	13	17	21	43	64	69	56	44	27	17	23
	PrecMax	663	32	35	37	68	152	178	148	149	84	95	43	40
	PrecMin	282	2	2	1	0	6	13	18	6	6	3	4	7
	PrecStd	87	8	8	10	16	37	41	36	33	22	21	12	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	2.9	8.9	21.4	28.4	30.1	30.0	31.7	29.3	21.4	10.6	4.1
		MCWB	1.7	1.4	5.1	10.7	14.3	18.8	22.1	19.0	17.2	13.1	6.7	2.2
	2%	DB	1.2	0.6	5.0	17.9	24.6	26.8	27.7	28.4	25.5	16.6	6.7	1.5
		MCWB	0.3	-0.3	2.9	9.3	13.0	17.2	20.4	18.8	15.5	10.0	4.1	0.2
	5%	DB	-1.1	-1.3	3.4	14.7	22.1	24.8	26.1	26.2	22.8	13.6	4.4	-1.0
		MCWB	-1.9	-1.9	1.8	8.1	12.0	16.1	19.3	18.2	14.7	8.7	2.6	-1.9
	10%	DB	-3.7	-3.3	1.9	11.4	19.8	22.8	24.4	24.1	20.0	11.0	2.3	-2.9
		MCWB	-4.2	-3.9	0.8	6.4	10.8	15.0	18.5	17.1	13.2	7.4	1.1	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	1.7	5.3	12.2	16.2	21.4	23.9	21.8	19.1	13.5	7.0	2.3
		MCDB	3.0	2.6	8.4	19.0	21.6	26.8	27.9	27.9	25.0	20.6	10.4	3.6
	2%	WB	0.3	-0.2	3.1	10.4	14.3	19.1	21.9	20.2	16.9	10.9	4.5	0.3
		MCDB	1.2	0.6	4.7	16.2	22.1	24.3	26.3	25.5	23.3	15.0	6.4	1.4
	5%	WB	-1.9	-1.9	1.9	8.5	13.0	17.6	20.4	19.1	15.2	9.1	2.7	-1.8
		MCDB	-1.2	-1.3	3.2	13.7	20.3	22.3	24.7	24.3	21.3	13.0	4.1	-1.0
	10%	WB	-4.2	-3.9	0.8	6.8	11.7	16.3	19.2	18.0	13.7	7.7	1.1	-3.4
		MCDB	-3.7	-3.3	1.9	11.2	18.4	20.7	23.1	22.8	19.1	10.5	2.3	-2.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	8.8	10.2	12.5	10.8	10.9	11.8	11.4	9.2	7.5	7.8
		MCDBR	9.4	8.6	8.4	15.2	15.8	14.1	13.4	15.0	15.8	13.8	9.6	9.2
		MCWBR	8.8	7.8	6.6	8.4	7.2	6.7	7.3	7.0	7.6	8.1	7.1	8.4
	5% WB	MCDBR	9.2	8.3	8.1	13.8	13.7	11.6	12.3	12.8	14.2	12.8	9.2	8.8
		MCWBR	8.7	7.6	6.6	8.4	6.9	6.7	7.7	7.1	7.4	7.8	7.1	8.1
Clear-Sky Solar Irradiance	taub		0.241	0.245	0.306	0.330	0.344	0.365	0.368	0.367	0.314	0.284	0.247	0.244
	taud		2.291	2.318	2.241	2.377	2.391	2.372	2.359	2.362	2.499	2.535	2.454	2.257
	Ebn at Noon		772	888	881	905	906	886	876	856	872	837	774	687
	Edh at Noon		59	78	107	107	111	114	114	108	83	64	50	51
All-Sky Solar Radiation	RadAvg		1.06	2.12	3.69	4.85	5.81	5.84	6.04	4.97	3.52	2.02	0.97	0.75
	RadStd		0.10	0.12	0.27	0.41	0.49	0.45	0.36	0.39	0.34	0.32	0.12	0.10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page